

BMIDE



Business Modeler IDE

What is BMIDE?

The Business Modeler IDE (Integrated Development Environment) is a tool for **configuring and extending the data model** of your Teamcenter installation. The data model objects define the objects and rules used in Teamcenter.

What is OOTB / COTS?

An out-of-the-box feature or functionality also called OOTB or off the shelf, is a native feature or built-in functionality of a product that comes directly from the vendor and works immediately when the product is placed in service.

OOTB and COTS are both terms that refer to pre-packaged software or applications that are ready to use and **don't require extensive customization or development**.

What is template in BMIDE?

In the Business Modeler IDE (BMIDE), a template is a set of XML files that contain the data model for an application. BMIDE users can create modifications to the data model using a graphical interface, which helps to avoid the complexity of writing XML files. The XML files are then combined into a custom template that Teamcenter uses to modify and extend the standard data model for the database.

Foundation_Template.xml file consist the data model for the base Teamcenter application.

What is the difference between hot deployment and cold deployment?

Hot deployment is deploying the changes directly in the production environment.

Cold deployment is first deploying the changes in the development environment, then testing environment and then production environment.

What is a Data Model?

A data model is a process and a way to create a custom business object, attach a list of values (LOV) to a property, and deploy the data model or a data model is a structured organization of abstract objects to represent business data.

Teamcenter has its own data model that you extend using the Business Modeler IDE.

Class

A class can be thought of as a storage class in the database. It is the persistent representation of an object table in the database, and each row in that table is an instance of the class.

Attributes

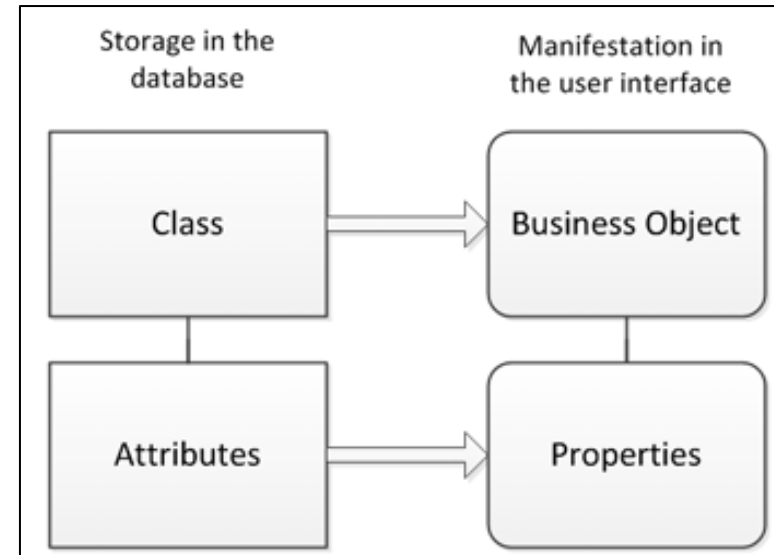
Attributes are the persistent characteristics of the class. Attributes of a class define the columns of the class table in the database.

Business Object

A business object represents a type of business data that you handle in Teamcenter client. Business object let you define the values to store in the database and set how objects behave, such as their list of values, naming rules, display name and so on.

Properties

Properties are business object characteristics.



The screenshot shows the SQL Server Enterprise Manager interface. On the left, the Object Explorer displays the database structure, with 'dbo.PDOCUMENTREVISION' selected. The main pane shows a SQL query in 'SQLQuery1.sql':

```
SELECT TOP (1000) [puid]
, [pDocumentTitle]
, [pDocumentAuthor]
, [pDocumentSubject]
FROM [DBUser1].[dbo].[PDOCUMENTREVISION]
```

Below the query, the 'Results' tab shows the output of the query:

	puid	pDocumentTitle	pDocumentAuthor	pDocumentSubject
1	A3DAAAbc5aSjxD	NULL	NULL	NULL
2	AmLAAA6h5aSjxD	NULL	NULL	NULL
3	QeAAAA7Y5aSjxD	NULL	NULL	NULL

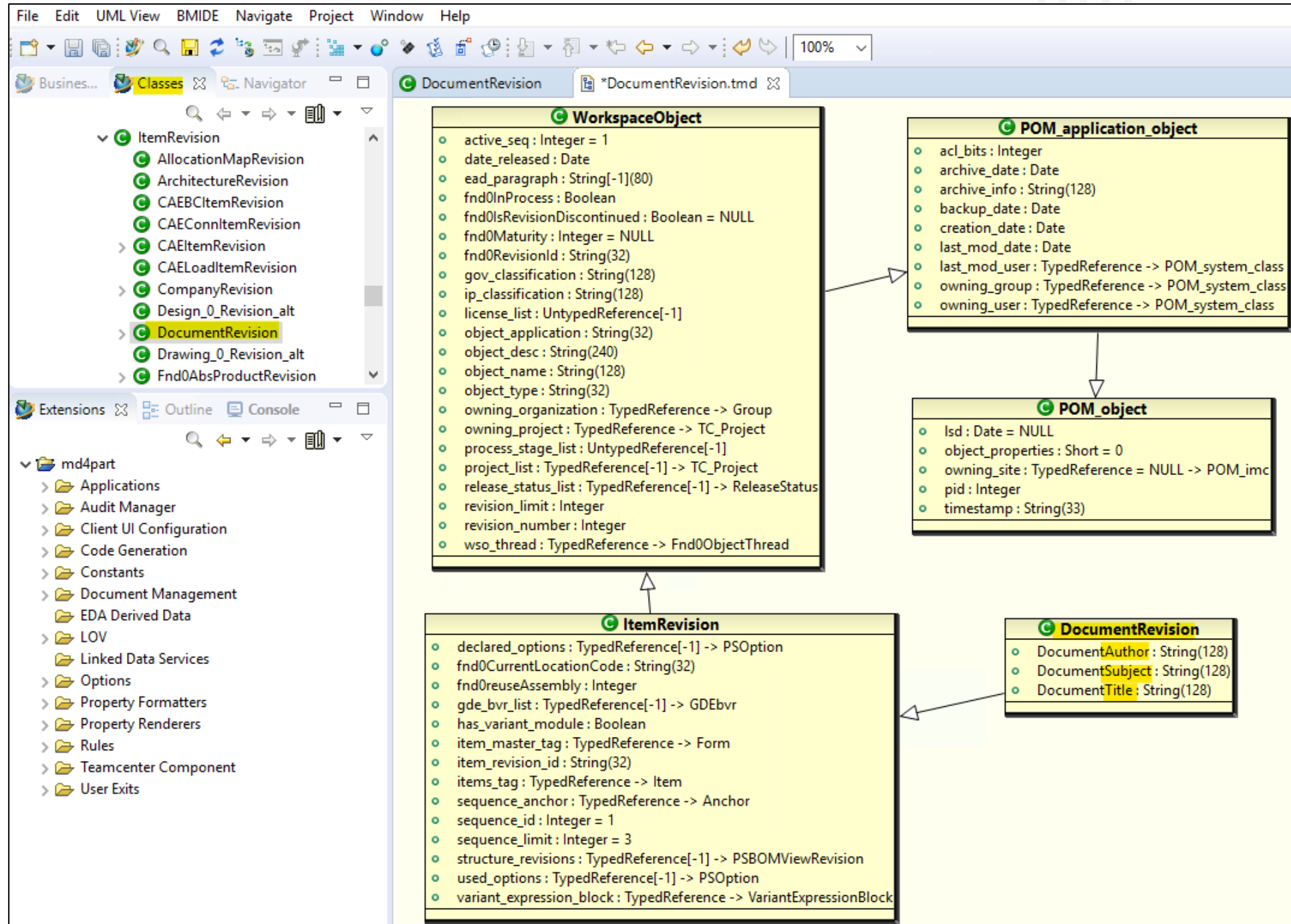
Image shows how data is stored in Database in means of Tables

- Initially table shows only 3 rows.
- We have create two new Documents in TC Rich Client and those items are get stored in database table in new rows .



Results					Messages				
	puid	pDocumentTitle	pDocumentAuthor	pDocumentSubject					
1	A3DAAAbc5aSjxD	NULL	NULL	NULL					
2	AeBAAArT5aSjxD	NULL	NULL	NULL					
3	AmLAAA6h5aSjxD	NULL	NULL	NULL					
4	QKEAAArT5aSjxD	Test	Test	Test					
5	QeAAAA7Y5aSjxD	NULL	NULL	NULL					

- The image shows, Document revision class has only 3 Attributes (Author, Subject, Title) and other Attributes are inherited from Parent.



Properties of Business Object

 New Property

Property Definition
Create New Property

Property Types

☒ **Persistent**

Persistent properties are properties of business objects that remain constant on the object.

☐ **Runtime**

Runtime properties are derived each time the property is displayed. Their data is derived from one or more pieces of system information (for example, date or time) that are not stored in the Teamcenter database.

☐ **Compound**

Compound properties are properties on business objects that can be displayed as properties of an object (the display object) although they are defined and reside on a different object (the source object).

☐ **Relation**

Relation properties are properties that define the relationship between objects. For example, a dataset can be attached to an item revision with a specification, requirement, or reference relation, among many others.

☐ **Table**

Table property provides the ability to manage an array of information/data in tabular manner. A full table property consists of rows. The number of rows can vary per business object instance.

☐ **Name-Value**

Name-Value property provides the ability to dynamically specify name-value pairs in a tabular format to represent ad-hoc object-specific characteristics not defined in the persistent properties for the business object.

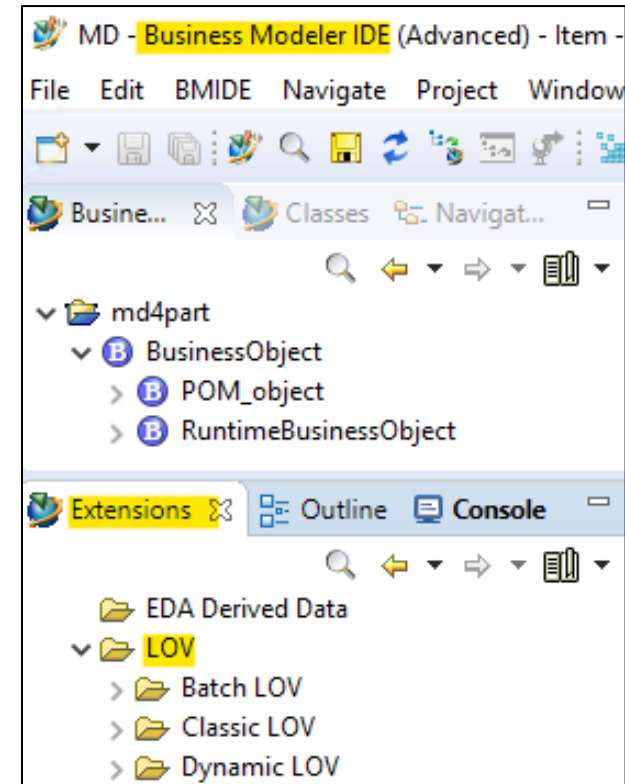
What is LOV?

LOV stands for List of Values, and it is a feature in Teamcenter BMIDE that allows users to select from a set of pre-defined values for a property on a business object.

LOVs are used to ensure that data is accurate and consistent by limiting the values that can be entered for a property.

There are three types of LOVs in Teamcenter:

1. **Classic LOV** : A list that is stored in the template
2. **Batch LOV** : A list that is stored in the Teamcenter database
3. **Dynamic LOV** : A list that queries the database at run time to show the valid values

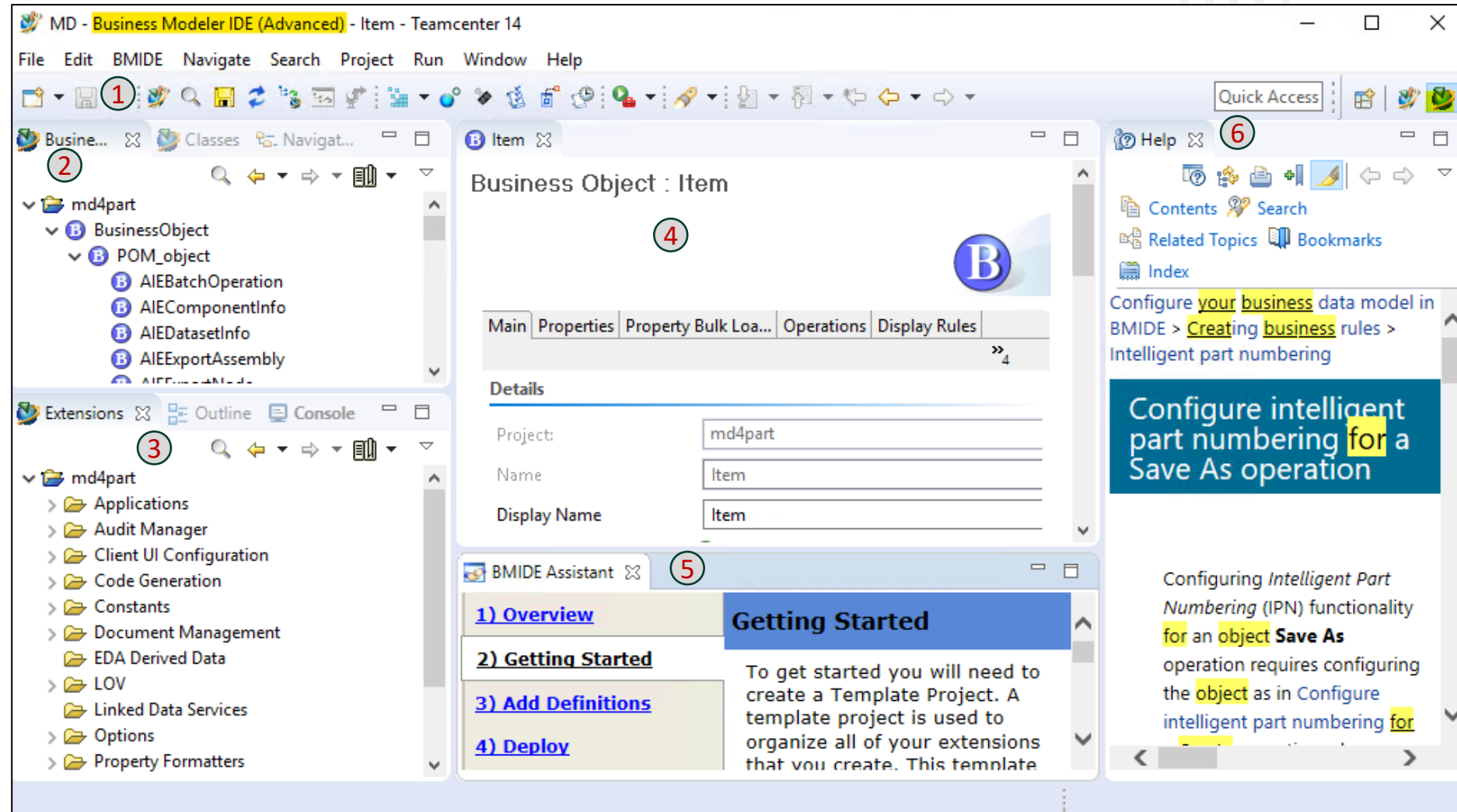


Business Modeler IDE interface

Advanced perspective

Provides a full-featured user interface. Contains separate views for business objects, classes, project files, and data model extensions

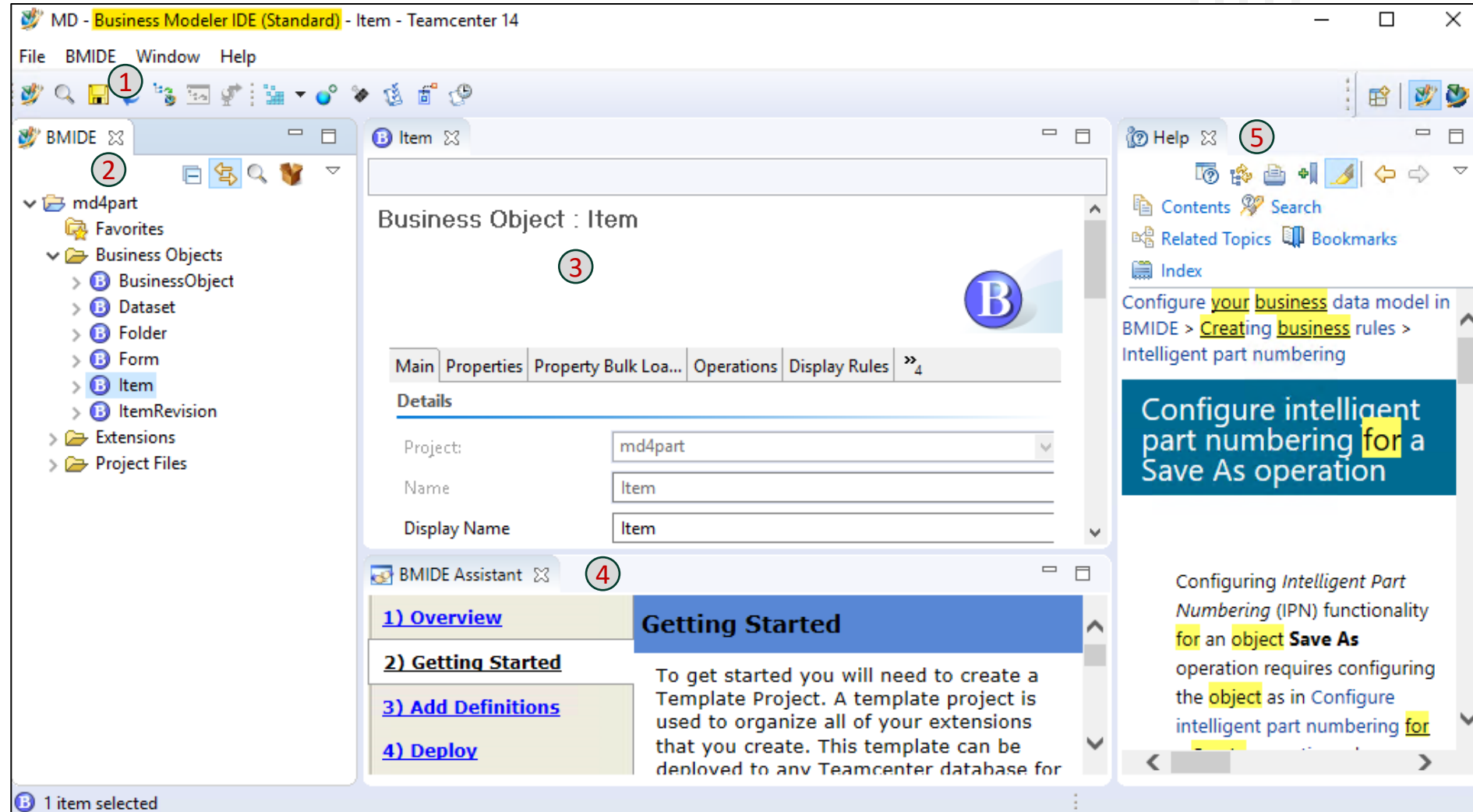
1. Toolbar
2. Business Object view
3. Extension View
4. Editor
5. BMIDE Assistant view
6. Help view



Standard perspective

Provides a simplified user interface. It contains the BMIDE view, which provides a centralized location for favorites, data model elements, and project files.

1. Toolbar
2. BMIDE View
3. Editor
4. BMIDE Assistant view
5. Help view



Create a BMIDE Template Project

- Choose File → New → Project → New Business Modeler IDE Template Project.
- Enter basic project information and click Finish.

New BMIDE Template Project

New Business Modeler IDE Template Project

Create a BMIDE Template project to organize all your extensions according to the Teamcenter Data Model, Behavior and Rules.

Prefix: * S4

Template name: * s4mytemplate

Template display name: * My template

Template description: * My Training Template

Dependent templates directory: * C:\SPLM\Teamcenter14\TR\bmide\templates Browse...

☒ Use default location

Location: C:\SPLM\Teamcenter14\TR\bmide\workspace\MD\s2mytemplate Browse...

Project "s2mytemplate" will be dependent upon "foundation" template by default.
To select more dependent templates. click Next.

< Back Next > **Finish** Cancel

New Project

Select a wizard

Create a new Business Modeler IDE Template Project

Wizards:

type filter text

- > General
- ▼ Business Modeler IDE
 - Create a new Teamcenter UML Diagram
 - New Business Modeler IDE Template Project**
 - New Composite Software Project
 - New Live Update Project

? < Back **Next >** Finish Cancel

New BMIDE Template Project

New Business Modeler IDE Template Project

✗ Invalid "Prefix:" field. Third party solution providers should use digits 2 or 3 in the prefix; Customers should use digits 4 through 9 in the prefix.

Prefix: * S1

New BMIDE Template Project

New Business Modeler IDE Template Project

✗ Invalid "Prefix:" field. The first character of the entry "prefix" must be an uppercase letter.

Prefix: * 11